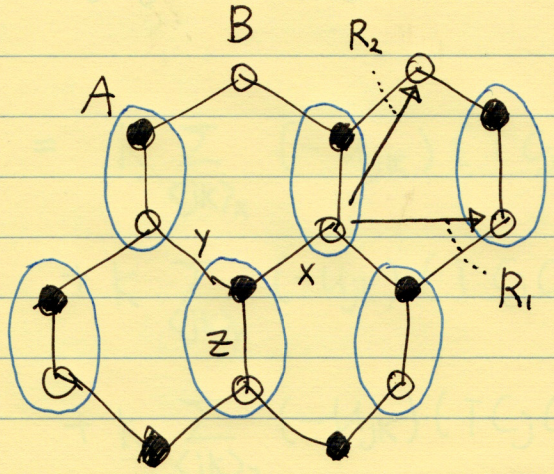
$$H = K \sum_{\langle jk \rangle_x} \sigma_j^x \sigma_k^x + K \sum_{\langle jk \rangle_y} \sigma_j^y \sigma_k^y + K \sum_{\langle jk \rangle_z} \sigma_j^z \sigma_k^z$$

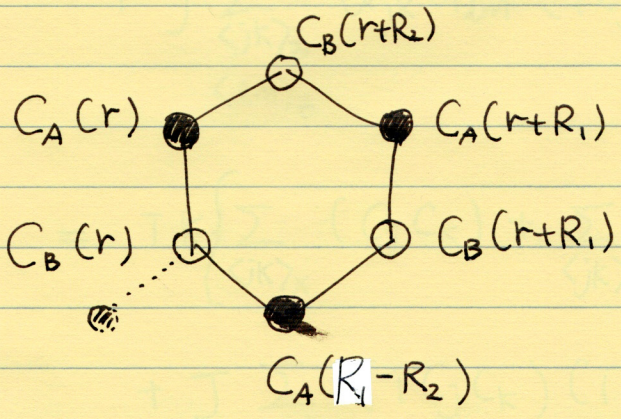
$$+ J \sum_{\substack{\langle jk \rangle_z \\ \langle lm \rangle_z}} \sigma_j^z \sigma_k^z \sigma_l^z \sigma_m^z$$



⬭ : z-bond

● : A sublattice

○ : B sublattice



푸리에 변환은

$$C_{\substack{A,\\B}}(r) = \frac{1}{\sqrt{N}} \sum_k e^{ik\cdot r} C_{\substack{A,\\B}}(k)$$

$$C_{\substack{A,\\B}}(r+R_1) = \frac{1}{\sqrt{N}} \sum_k e^{ik\cdot(r+R_1)} C_{\substack{A,\\B}}(k)$$

이때, flux sector를 $W_p = +1$ 에 대해서

$u_{jk} = -1$ (j : A sublattice
k : B sublattice
즉, $u_{AB} = -1$).

를 gauge configuration 으로 적용하면.

$$H = K \sum_{\langle jk \rangle_x} (-u_{jk})(\bar{\imath} c_j c_k)$$
$$+ K \sum_{\langle jk \rangle_y} (-u_{jk})(\bar{\imath} c_j c_k)$$
$$+ K \sum_{\langle jk \rangle_z} (-u_{jk})(\bar{\imath} c_j c_k)$$
$$+ J \sum_{\substack{\langle jk \rangle_z \\ \langle lm \rangle_z}} u_{jk} u_{lm} (\bar{\imath} c_j c_k)(\bar{\imath} c_l c_m)$$
$$= \bar{\imath} K \left\{ \sum_{\langle jk \rangle_x} (c_j c_k) + \sum_{\langle jk \rangle_y} (c_j c_k) + \sum_{\langle jk \rangle_z} (c_j c_k) \right\}$$
$$+ J \sum_{\substack{\langle jk \rangle_z \\ \langle lm \rangle_z}} (\bar{\imath} c_j c_k)(\bar{\imath} c_l c_m)$$

이다.

이때, Mean-field decoupling 은 두 연산자 A, B에 대해서

$$(A-\langle A\rangle)(B-\langle B\rangle) \cong 0 \quad \text{을 가정하여}$$

$$\Rightarrow AB \cong A\langle B\rangle + \langle A\rangle B - \langle A\rangle\langle B\rangle$$

임을 이용하는 것이다. 우리의 상황에서 $C_jC_kC_\ell C_m$ 에 이 MFT 를 적용해보면

$$C_jC_kC_\ell C_m = \frac{1}{3}\Big[\overbrace{C_jC_k}^{A}\overbrace{C_\ell C_m}^{B} - \overbrace{C_jC_\ell}^{A}\overbrace{C_kC_m}^{B} + \underbrace{C_jC_m}_{A}\underbrace{C_kC_\ell}_{B} \Big]$$

(*Majoran fermion operator의 반교환 관계)

$$\text{(MFT)} \rightarrow \cong \frac{1}{3}\Big[\{\langle C_jC_k\rangle C_\ell C_m + \langle C_\ell C_m\rangle C_jC_k - \langle C_jC_k\rangle\langle C_\ell C_m\rangle\}$$

$$(-1)\{\langle C_jC_\ell\rangle C_kC_m + \langle C_kC_m\rangle C_jC_\ell - \langle C_jC_\ell\rangle C_kC_m\}$$

$$+\{\langle C_jC_m\rangle C_kC_\ell + \langle C_kC_\ell\rangle C_jC_m - \langle C_jC_m\rangle\langle C_kC_\ell\rangle\}\Big]$$

이다. 이때, 우리는 우선 간단한 상황을 다루기 위해서 첫 번째 중괄호에 해당하는 직관적인 term

$$\langle C_jC_k\rangle C_\ell C_m + \langle C_\ell C_m\rangle C_jC_k - \langle C_jC_k\rangle\langle C_\ell C_m\rangle$$

만을 보고, 이때 $\langle jk\rangle_z$, $\langle \ell m\rangle_z$ 의 "z-bond" 에 해당하는 $(jk\ell m)$ 의 configuration 만을 우선 고려한다.

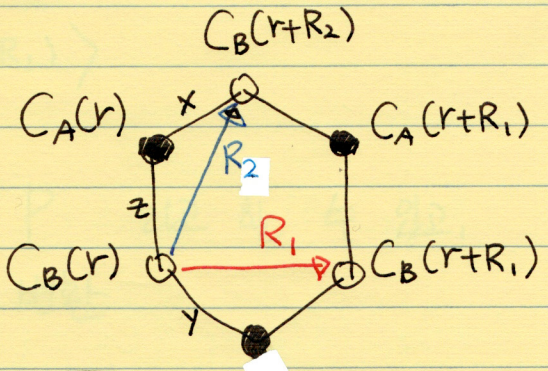
그렇게 되면 기존의 Standard Kitaev model 에서
z-bond 가 강화되거나 약화되는 효과를 기대할 수 있을 것이다.

그러면 이제, 푸리에 변환을 통해서 $C_S(r)$ 들을 $C_S(k)$ 들로 바꿔서 해밀토니안을 작성해보면,

$$H = iK \sum_{\langle jk\rangle_x} c_j c_k$$

$$+ iK \sum_{\langle jk\rangle_y} c_j c_k$$

$$+ iK \sum_{\langle jk\rangle_z} c_j c_k$$



$u_{AB} = -1$

$$+ J \sum_{\substack{\langle jk\rangle_z \\ \langle lm\rangle_z}} \left[\langle i c_j c_k \rangle c_l c_m + \langle i c_l c_m \rangle c_j c_k + \langle i c_j c_k \rangle \langle i c_l c_m \rangle \right]$$

$\sum_r$ 은 H를 구성할 때, 모든 (무한한 격자) 유닛셀에 대한 합의 해당.

$$\simeq iK \sum_r C_A(r)\, C_B(r+R_2) \quad \cdots \text{ x-bond , k-coupling}$$

$$+ iK \sum_r C_A(r+R_1)\, C_B(r+R_2) \quad \cdots \text{ y-bond , k-coupling}$$

$$+ iK \sum_r C_A(r)\, C_B(r) \quad \cdots \text{ z-bond , k-coupling}$$

$$+ J \sum_r \Big[\langle i C_A(r) C_B(r) \rangle C_A(r+R_1) C_B(r+R_1)$$

$$+ \langle i C_A(r+R_1) C_B(r+R_1) \rangle C_A(r) C_B(r)$$

$$+ \langle i C_A(r) C_B(r) \rangle \langle i C_A(r+R_1) C_B(r+R_1) \rangle \Big]$$

이때, mean-field parameter $P_1 \equiv \langle \bar{T} C_A(r) C_B(r) \rangle$

를 정의하고, translational symmetry 가 있다고 가정하면,

$$P_1 \equiv \langle \bar{T} C_A(r) C_B(r) \rangle$$

$$P_2 \equiv \langle \bar{T} C_A(r+R_1) C_B(r+R_1) \rangle$$

에 대해서, $P_1 = P_2 = P$ 라고 할 수 있고,

따라서 해밀토니안은 푸리에 변환

$$\sum_r C_A(r+R_1) C_B(r+R_1)$$

$$= \sum_r \left(\frac{1}{\sqrt{N}} \sum_k e^{\bar{T} k \cdot (r+R_1)} C_A(k) \right) \left(\frac{1}{\sqrt{N}} \sum_{k'} e^{\bar{T} k' \cdot (r+R_2)} C_B(k') \right)$$

$$= \frac{1}{N} \sum_k \underbrace{\sum_r e^{\bar{T}(k+k') \cdot r}}_{= N \cdot \delta_{k',-k}} e^{\bar{T} k \cdot R_1} e^{\bar{T} k' \cdot R_2} C_A(k) C_B(k')$$

$$= \sum_k e^{\bar{T} k \cdot (R_1 - R_2)} C_A(k) C_B(-k)$$

$$= \sum_k e^{\bar{T} k \cdot (R_1 - R_2)} C_A(k) C_B^{\dagger}(k)$$

등을 이용하면,

$$H(p) \cong H_{MF}(p)$$

$$H_{MF}(p) = \bar{1}\sum_k \left\{ K\left(1 + e^{-\bar{1}k\cdot R_2} + e^{\bar{1}k\cdot(R_1-R_2)}\right) + 2Jp \right\} C_A(k)\, C_B(-k)$$

$$- \sum_r Jp^2$$

1BZ 에서는 $\sum_k (\cdots) = \sum_{-k} (\cdots)$ (k를 바꿈.) 이다.

$$= \frac{\bar{1}}{2}\sum_k \left\{ K\left(1 + e^{-\bar{1}k\cdot R_2} + e^{\bar{1}k\cdot(R_1-R_2)}\right) + 2Jp \right\} C_A(k)\, C_B(-k)$$

$$- \frac{\bar{1}}{2}\sum_k \left\{ K\left(1 + e^{\bar{1}k\cdot R_2} + e^{-\bar{1}k\cdot(R_1-R_2)}\right) + 2Jp \right\} C_B(k)\, C_A(-k)$$

$$- \sum_r Jp^2$$

$$D(k) \equiv K\left(1 + e^{\bar{1}k\cdot R_2} + e^{-\bar{1}k\cdot(R_1-R_2)}\right) + 2Jp$$

$$= \frac{\bar{1}}{2}\sum_k \left\{ K\left(1 + e^{\bar{1}k\cdot R_2} + e^{-\bar{1}k\cdot(R_1-R_2)}\right) + 2Jp \right\} C_A(-k)\, C_B(k)$$

$$- \frac{\bar{1}}{2}\sum_k \left\{ K\left(1 + e^{-\bar{1}k\cdot R_2} + e^{\bar{1}k\cdot(R_1-R_2)}\right) + 2Jp \right\} C_B(-k)\, C_A(k)$$

$$- \sum_r Jp^2$$

$$\equiv \tilde{H}_{MF}(k;p) = U^{\dagger}(k) \begin{pmatrix} \omega_k & 0 \\ 0 & -\omega_k \end{pmatrix} U(k)$$

$$= \frac{1}{2}\sum_k \begin{pmatrix} C_A^{\dagger}(k) & C_B^{\dagger}(k) \end{pmatrix} \begin{pmatrix} 0 & \bar{1}D(k) \\ -\bar{1}D^{*}(k) & 0 \end{pmatrix} \begin{pmatrix} C_A(k) \\ C_B(k) \end{pmatrix} - \sum_r Jp^2$$

이고,

$$\begin{cases} D(k) = K\left(1 + e^{\bar{1}k\cdot R_2} + e^{-\bar{1}k\cdot(R_1-R_2)}\right) + 2Jp \\ \tilde{H}_{MF}(k) = U(k) \begin{pmatrix} \omega_k & 0 \\ 0 & -\omega_k \end{pmatrix} U_k \end{cases}$$

일때, $\omega_k = |D(k)|$ 이다.